

closure_evidence_20260922

BOB-209 closure evidence — 2026-09-22

Fix (three defects, one function, scripts/lib/ ownership.sh probe_location())

1. **Asymmetric guards:** both the file branch and the directory/probe branch now check stat's exit code AND output emptiness together (previously each checked only one half).
2. **Mislabelled verdict:** a stat failure (either branch) now echoes stat-failed, never the semantically-wrong unwritable (which stays reserved for the case mktemp itself could not create the probe — a genuine writability finding).
3. **Untrapped cleanup (§11.4.14):** cleanup moved from a bare `rm -f` statement to a RETURN trap (covers every return path uniformly) plus saved/restored INT/TERM/HUP handlers — removes the probe first, restores whatever handler the caller already had (ownership_repair.sh installs its own top-level INT/TERM handlers), then re-raises so the caller's own handling still runs exactly as before.

RED/GREEN evidence (via byte-identical pre- fix source run under the identical harness)

- File-branch empty-output: pre-fix wrong-owner: (malformed) → post-fix stat-failed.
- File-branch stat failure: pre-fix unwritable (mislabel) → post-fix stat-failed.
- Dir-branch stat failure leaking garbage output: pre-fix trusted the garbage as a uid (wrong-owner:999999) → post-fix stat-failed.

- Interrupt-between-mktemp-and-cleanup (a real SIGINT, deterministically timed via a shimmed stat that backgrounds the real stat, writes a checkpoint file, then sleeps; the harness polls for the checkpoint then signals): pre-fix strands a .ownership-probe.* file; post-fix leaves none.

Independently re-verified this session

```
$ bash tests/unit/test_ownership_probe_location.sh  
RESULT: 10 passed, 0 failed, 0 skipped
```

Plus the full tests/unit/test_ownership_repair.sh (146/146) and tests/pre_build/test_check_cm_ownership_invariants.sh (42/42).

See docs/qa/B0B-208/closure_evidence_20260922.md for the combined
git diff --stat.